

Project Name Suggestions

- **GestureDrive: Computer Vision-Based Real-Time Hand Gesture Control System for Virtual Vehicles**
- **AirSteer: Interactive 3D Driving Simulator Powered by MediaPipe Hand Tracking**
- **VisionMotion: Touchless Biometric Control Interface for 3D Automotive Simulation**

Science Fair Presentation Script

1. Introduction (30 seconds)

"Good morning/afternoon, judges. My name is [Your Name], and my project is **GestureDrive**. Have you ever wondered how we can interact with digital environments without physical controllers, keyboards, or mice? For my project, I built a real-time, browser-based 3D driving simulator that allows a user to fully control a supercar—accelerating, braking, steering, shifting gears, and toggling headlights—using **only bare-hand gestures** captured through a standard webcam."

2. The Problem & Motivation (45 seconds)

"Traditional video games and simulators rely heavily on physical hardware like gamepads, steering wheels, or keyboards. While effective, these devices lack intuitive, touchless interaction models. My goal was to explore the practical application of **computer vision and machine learning** in human-computer interaction (HCI). I wanted to answer the question: *Can a standard webcam accurately translate human hand landmarks into low-latency physics controls for a 3D application?*"

3. Methodology & Technologies Used (1 minute)

"To achieve this, I integrated several powerful web technologies:

- **Three.js:** Used to build the 3D graphics pipeline, rendering the supercar, dynamic road environments, lighting effects, and shadows entirely in the browser using WebGL.
- **Google MediaPipe Hands:** A lightweight machine learning framework that detects 21 3D landmarks on a human hand per frame.
- **Custom Gesture Logic:** I wrote algorithms on top of MediaPipe's landmark data to map specific hand states—such as finger extension ratios, hand openness, and relative wrist positioning—into driving commands. For instance, closing your right hand accelerates the car, opening your left hand applies the brakes, and using a two-finger gesture toggles the headlights."

4. Live Demonstration (45 seconds)

"Let me show you a quick demonstration. As you can see, the webcam feed in the corner tracks my hand skeleton in real-time. When I close my right fist, the car accelerates. If I tilt my hands left or right, the steering angle updates instantly. By opening both hands, the vehicle shifts into reverse. If an obstacle appears, I can brake instantly by opening my left hand."

5. Conclusion & Future Scope (30 seconds)

"In conclusion, **GestureDrive** proves that high-performance, markerless gesture

recognition can be successfully paired with real-time 3D physics engines using affordable hardware like a standard laptop webcam. In the future, this technology could be expanded into assistive driving interfaces, touchless smart-home dashboards, or virtual reality navigation tools. Thank you, and I am now open to any questions."

Practice Questions from Judges & Sample Answers

Q1: "How does the system actually track your hands? What library are you using?"

Answer: "The application uses **Google MediaPipe Hands**, a machine learning pipeline running locally in the browser via JavaScript. It detects 21 3D coordinates (landmarks) on each hand from the live video stream. My custom JavaScript script then analyzes the Euclidean distances between the fingertips and the wrist base to determine if a hand is open, closed, or showing a specific number of fingers."

Q2: "What happens if the lighting is poor or the webcam fails to detect your hand?"

Answer: "Lighting can impact computer vision accuracy. If MediaPipe loses track of the hand, the application features a **built-in fallback control system**. Users can seamlessly switch to keyboard controls using the arrow keys for steering, 'W' and 'S' for acceleration and braking, and 'R' for reverse."

Q3: "How do you handle latency? Isn't real-time rendering and machine learning heavy for a web browser?"

Answer: "Latency is minimized by running optimized WebGL rendering via **Three.js** and configuring MediaPipe with a balanced model complexity. The hand-tracking loop and the 3D render loop are synchronized using requestAnimationFrame, ensuring a smooth frame rate without noticeable input lag."

Q4: "What were the biggest challenges you faced while building this project?"

Answer: "One of the biggest challenges was **state management for gear shifting and gestures**. Initially, transitioning from acceleration to braking caused jitter because hand states flickered between open and closed frames. To fix this, I implemented custom cooldown timers and threshold multipliers on the finger extension ratios to ensure reliable, debounced input recognition."

Q5: "How could this project be applied in the real world outside of a science fair simulation?"

Answer: "While this is a driving simulator, the underlying computer vision framework has major applications in **human-computer interaction**. It can be used for touchless kiosk interfaces in public spaces for hygiene, automotive cabin controls where drivers can adjust music or climate without looking away from the road, or as an accessibility tool for individuals with limited mobility who struggle with traditional controllers."